

CHAPTER 2

Life on Earth

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.
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The Big Questions

1 How do physiography and climate shape life on Earth?

✔ ANSWER

- Temperature, rainfall and landforms decide which plants and animals can live in a place.
- So the Earth is divided into biomes: forests, grasslands, deserts, tundra and aquatic biomes.
- Hot, wet places near the Equator have dense rainforests; dry places have deserts with cacti and camels; the cold tundra has only mosses, lichens and small shrubs.
- Humans also adapt their homes, food and work to these conditions.

2 How are humans and other life forms interconnected?

✔ ANSWER

- Biomes give plants, animals and humans food, water, shelter and resources.
- Living beings also change biomes: animals help in pollination and seed dispersal, microorganisms recycle nutrients, humans change land for farms and cities.
- Forest dwellers depend on forests for food, fuel, fodder, medicine and income, and also protect them.

- Together they keep ecological balance and biodiversity.

3 **What is the importance of preserving various life forms, and how can we contribute to their conservation?**

✔ ANSWER

- Biodiversity keeps ecosystems healthy, gives us food, medicine and resources, and balances oxygen and climate.
- We can help by protecting hotspots and protected areas, planting trees, preventing forest fires, saving water bodies and commons, practising ecotourism, and supporting forest communities (e.g. Mendha Lekha, sacred groves).

⭐ Let's Map · Biomes of the world (Fig. 2.2)

1 **Which type of biome is found in smaller, separate regions, and why?**

💡 SAMPLE ANSWER

The **tundra biome** is found only in small, separate patches along the Arctic coasts and on high mountains, because very cold conditions are found only in these narrow areas.

(Tropical rainforests are also in separate blocks — Amazon, Congo, Southeast Asia — because they need hot and wet conditions found only near the Equator.) (check the map in your book)

2 **Which continent has the least number of biomes? What could be the reason for this?**

💡 **SAMPLE ANSWER**

Antarctica — it is almost fully covered with ice and is extremely cold, so no biome with plants and animals is shown there.

Among the other continents, **Australia** has few biomes (mostly desert and grassland) because a large part of it is dry. (check the map in your book)

3 **Why are there areas that do not have any specific biomes?**

💡 **SAMPLE ANSWER**

Some areas, like Greenland, Antarctica and very high mountains, are covered with ice and snow all year. It is too cold for plants to grow, so they are not shown as any biome.

⭐ Let's Explore · Forest Biome

1 **Look at the layers of vegetation in a rainforest (Fig. 2.4). What do the yellow shapes in the figure indicate?**

✔ ANSWER

The yellow shapes show **how much sunlight reaches each layer**.

- Emergent layer: full circle — gets the most sunlight.
- Upper canopy and understory: less and less sunlight.
- Forest floor: only a small part — very little sunlight reaches it because of the dense trees above.

⭐ Let's Explore · Grassland Biome

1 **Observe Fig. 2.5 and explore which of these animals you will find in the grasslands of India.**

💡 SAMPLE ANSWER

Fig. 2.5 shows an elephant, lions, antelopes/deer, a giraffe and a zebra.

- Found in India: **elephants, lions** (Gir in Gujarat) and **antelopes/deer** (like blackbuck and chinkara).

- Not found in the wild in India: **giraffe and zebra** (they live in the African savannas).

⭐ Let's Explore · Desert Biome

1 **What types of activities do you think people in desert regions generally undertake to sustain their households and livelihood?**

💡 SAMPLE ANSWER

- Rearing animals like camels, goats and sheep (herding).
- Growing crops that need little water (bajra, pulses).
- Handicrafts, weaving, embroidery and pottery.
- Trade, transport on camels, salt making (Rann of Kutch), mining and tourism.

2 **With the help of your teacher, explore how people address issues related to the availability of water.**

💡 SAMPLE ANSWER

- They collect rainwater in tanks, ponds and underground structures (like kunds/tankas).
- They build step-wells (baoris) and village ponds (johads).

- They use water carefully, carry it from far sources (as in Fig. 2.6), and use canals and drip irrigation.

3 **Compare the physical features of the Rajasthan camel (Fig. 2.7a) and the Nubra Valley (Ladakh) camel (Fig. 2.7b).**

💡 SAMPLE ANSWER

Rajasthan camel	Nubra Valley camel
One hump	Two humps
Short, thin hair	Thick, long, woolly fur
Tall with long legs	Shorter and stockier body

4 **Considering the climatic conditions, why do you think the Nubra Valley camel might have a different fur density than the Rajasthan camel?**

✔ ANSWER

Nubra Valley in Ladakh is a **cold desert** with freezing winters, so the camel needs thick fur to keep warm.

Rajasthan's Thar is a **hot desert**, so the camel has thin, short hair that helps it stay cool.

🤔 Think About It · Tundra Biome

1 **Tundra in polar regions has mosses, lichens, shrubs, etc., while the Himalaya, in the subtropics, is dominated by conifers.**

What could be the reason for this? What difference do you find in the animals of these two regions? Which species are found only in the Himalaya?

💡 SAMPLE ANSWER

- **Reason:** The tundra is the coldest biome with permafrost (frozen soil) and very short summers, so trees cannot grow. The Himalaya lies in the subtropics, gets more sunlight, a longer growing season and more rainfall on its slopes, so conifers (pines, firs) can grow up to a height.
- **Animals:** The tundra has reindeer, arctic foxes and polar bears. The Himalaya has snow leopards, yaks, Himalayan tahr, musk deer, red panda and the Himalayan snowcock.
- **Only in the Himalaya:** e.g. Himalayan yak, snow leopard, Himalayan snowcock, red panda.

⭐ Let's Explore · Aquatic Biomes

1 **Explore your local environment and complete the table: Component type – Examples**

observed – Interaction with humans.

💡 SAMPLE ANSWER

Component type	Examples observed	Interaction with humans
Trees/Vegetation/Plants	Neem, peepal, mango, grass	Shade, fruit, wood, cleaner air
Insects	Bees, butterflies, ants, mosquitoes	Pollination, honey; mosquitoes spread disease
Animals	Cows, dogs, birds, monkeys	Milk, farm work; monkeys damage crops
Non-living things	Soil, water, air, sunlight	Farming, drinking water, energy

(Fill in what you see in your own area.)

⭐ Let's Map · Biome Pyramid (Fig. 2.10)

1 **Mark the following regions at appropriate levels on Fig. 2.10: Andaman & Nicobar Islands, Keralam, Ladakh, Chhattisgarh, Odisha, Meghalaya, and Lakshadweep. Where will you place your home?**

💡 SAMPLE ANSWER

- Ladakh: top (cold) level — **1 Tundra** (cold, dry).
- Andaman & Nicobar Islands, Meghalaya, Keralam: tropical, wet — **7 Rainforest**.
- Lakshadweep: tropical, wet coast — near **7/8**.
- Chhattisgarh, Odisha: tropical — **9 Tropical Seasonal Forest** (deciduous forests).
- Your home: place it by your area's temperature and rainfall (e.g. Delhi: tropical, drier — near 10 Savanna/11 Desert).

2 **There is no arrow on the third side of the triangle. What would happen if we drew an arrow? Would it point upwards or downwards?**

💡 SAMPLE ANSWER

The right arrow shows cold → hot and the bottom arrow shows wet → dry.

An arrow on the left side would go from the cold polar top down to the hot, wet tropics. It would point **downwards**, showing that the number of species, plant growth and biodiversity **increase** as we move towards warm, wet tropical regions (the base is widest).

⭐ Let's Map · Biodiversity regions (Fig. 2.13)

1 **Mark and name on the map of India: 1. Namdapha National Park 2. Rann of Kutch 3. Gahirmatha beach 4. Gulf of Mannar 5. Pachmarhi 6. Achanakmar–Amarkantak. Gather information and discuss their features.**

💡 SAMPLE ANSWER

- **Namdapha National Park** – Arunachal Pradesh (far northeast); dense tropical rainforest.
- **Rann of Kutch** – Gujarat; one of the world's largest salt deserts, a marshy wetland in the monsoon.
- **Gahirmatha beach** – Odisha coast; nesting site of Olive Ridley sea turtles.
- **Gulf of Mannar** – between Tamil Nadu and Sri Lanka; marine reserve with coral reefs.
- **Pachmarhi** – Madhya Pradesh (Satpura hills); biosphere reserve.

- **Achanakmar–Amarkantak** – Madhya Pradesh–Chhattisgarh border; biosphere reserve, source of the Narmada.

⭐ Let's Explore · Forest Ecotourism (Thenmala)

1 **Is there any such ecotourism site in your State/UT? Collect some information and discuss with your friends.**

💡 SAMPLE ANSWER

Example: **Kanha National Park (Madhya Pradesh)** — visitors go on guided jungle safaris, local people work as guides and run homestays, and entry fees help protect tigers and forests.

(Write about a site in your own State/UT.)

2 **What should be a tourist's behaviour while visiting any tourist site? What should one be mindful of?**

💡 SAMPLE ANSWER

- Do not litter; carry back plastic and waste.
- Do not disturb, feed or tease animals; keep noise low.
- Stay on marked paths and follow rules and guides.
- Do not pluck plants or damage monuments.

- Respect local people and their culture; buy local products.

🤔 Think About It · Sacred Groves

1 **Sacred groves are small forest patches protected by local communities (Devrai, Devarakadu, Kavu, Sarna). India has over 1,00,000 of them.**

Why are these groves so special?

✔ ANSWER

- Though small, they are **rich biodiversity hotspots**.
- They preserve rare plants, medicinal species and wildlife that may not survive elsewhere.
- Some are linked to water sources and help maintain streams and groundwater.
- They show how faith and culture help local communities protect nature.

⭐ Let's Explore · Forest Dwellers

1 **Try to identify and classify the items for sale in the images given in Fig. 2.18. Also, make a list of the food and medicinal varieties available in your local market.**

💡 SAMPLE ANSWER

Items in the pictures (check the picture in your book):

- Handicrafts: bamboo baskets.
- Food: wild fruits/forest produce, mushrooms.
- Medicinal: roots and dried herbs.

Local market list (example): food — honey, mushrooms, amla, jamun, bamboo shoots; medicinal — tulsi, neem, ashwagandha, turmeric, giloy.

⭐ Let's Explore · GI-tagged Forest Products

1 **Can you explain how GI tagging of the forest products mentioned above connects geography, cultural traditions, local governance, and rural economy?**

✔ ANSWER

- **Geography:** the product's quality comes from the place — forest soil and climate (e.g. Tezpur Litchi, Nilgiri Tea).
- **Culture:** it uses traditional knowledge and crafts (e.g. Channapatna toys, Kandhamal Haldi grown by tribal communities).
- **Local governance:** local groups and communities register and protect the product's name.
- **Rural economy:** exclusive market recognition brings better prices and income to local people.

2 **Can you name some GI-tagged forest products from your state or union territory?**

💡 SAMPLE ANSWER

Examples from the chapter: Ramban Sulai Honey (J&K), Tezpur Litchi (Assam), Channapatna Toys (Karnataka), Kandhamal Haldi (Odisha), Erode Turmeric (Tamil Nadu), Nilgiri Tea, Tripura Queen Pineapple (Tripura), Khasi Mandarin (Meghalaya).

(Name the ones from your own State/UT; see the School Bhuvan–NCERT Geoportal.)

🤔 Think About It · Forest Fires

1 **According to the Forest Survey of India (FSI), about 36 per cent of India's forest cover is prone to forest fires, nearly 4 per cent extremely prone.**

Climate change and human-induced factors are the main reasons. (Think about the causes and effects.)

💡 SAMPLE ANSWER

- **Causes:** hot, dry weather due to climate change; careless burning of waste or crop residue, campfires, cigarettes, clearing land by fire.
- **Effects:** loss of human life, homes and resources; loss of trees and wildlife; smoke and pollution.

- **Prevention:** human care — no fires in forests, fire lines, awareness and early warning.

⭐ Let's Explore · Pradhan Mantri Ujjwala Yojana

1 **You may have heard about the Pradhan Mantri Ujjwala Yojana (PMUY). How are the benefits of this scheme impacting the health of women?**

💡 SAMPLE ANSWER

PMUY gives LPG cooking gas connections to poor households.

- Women no longer breathe smoke from firewood and cow dung, so eye, lung and breathing problems reduce.
- They save time and effort spent collecting fuelwood.
- Less cutting of wood also helps forests.

🤔 Think About It · Human–Wildlife Conflict

1 **Which wild animals (like monkeys or deer) have you seen in your residential area?**

💡 SAMPLE ANSWER

Example: monkeys, peacocks, squirrels, snakes, jackals, nilgai or wild boar (write what you have seen).

2 **Identify the causes behind these animals' presence in human settlements.**

💡 SAMPLE ANSWER

- Forests are being cut, so animals lose their homes (habitat loss).
- Lack of food and water in forests.
- Easy food in towns: crops, garbage, fruit trees.
- Roads and settlements built through forests and animal paths.

❓ Questions and Activities

1 **Why does India have diverse biomes? How does the local geography connect to the diversity?**

✔ ANSWER

India has very varied landforms, climate and rainfall:

- The cold Himalaya and Ladakh (cold desert, conifers).
- The hot, dry Thar Desert and the salt desert of the Rann of Kutch.
- Wet tropical rainforests in the Northeast, Western Ghats and Andaman & Nicobar Islands.
- Grasslands, deciduous forests, long coasts, rivers, lakes and coral reefs.

Different temperature, rainfall and relief in each area support different plants and animals, so India has many biomes.

2 **Make a list of economically valuable native plants in your region and comment on their abundance.**

💡 **SAMPLE ANSWER**

Example list: neem (common), bamboo (common in forests), sal and teak (decreasing due to cutting), mahua and tendu (found in forest areas), tulsi and ashwagandha (grown in gardens).

Comment: many useful plants are reducing because of overuse and loss of forests, so they must be planted and used sustainably.

3 **Prepare a map of your state or Union Territory showing national parks and sanctuaries.**

💡 **SAMPLE ANSWER**

Steps: take an outline map of your State/UT; mark the national parks and wildlife sanctuaries with symbols; add a key, title, north arrow and scale.

Example (Madhya Pradesh): Kanha, Bandhavgarh, Pench, Satpura and Panna National Parks.

Use the School Bhuvan–NCERT Geoportal for help.

4 **Study the impact of forest fires on people dependent on forests. Try to include the causes, losses, and preventive measures.**

💡 SAMPLE ANSWER

- **Causes:** dry, hot weather and climate change; human carelessness (campfires, cigarettes, burning waste, clearing land).
- **Losses:** homes and resources destroyed; loss of firewood, honey, bamboo, medicinal plants and income; animals die; smoke harms health.
- **Prevention:** no open fires in forests; clearing dry leaves and making fire lines; watch towers and early warning; training local volunteers; quick reporting to the forest department.

5 **Why are forests often called the 'lungs of the Earth'?**

✔ ANSWER

Like lungs, forests help us breathe:

- Trees take in carbon dioxide and give out oxygen through photosynthesis.
- They maintain oxygen levels and help balance the Earth's climate.

6 **If large forest areas continue to be destroyed, what do you think would be its long-term effect on the global climate?**

💡 **SAMPLE ANSWER**

- More carbon dioxide in the air, so global warming increases.
- Rainfall patterns change; more droughts and floods.
- Loss of biodiversity and gene banks.
- More soil erosion, landslides and forest fires.

7 **Discuss some traditional forest conservation methods with your grandparents or an elderly person.**

💡 **SAMPLE ANSWER**

Sample points:

- Protecting sacred groves (Devrai, Kavu, Sarna) where trees are never cut.
- Rituals to honour forest deities before collecting resources (Gonds, Bhils).
- Taking only what is needed; cutting only dry branches for firewood.
- Movements like the Bishnoi Movement and Chipko Movement to save trees.
- Community rules on grazing and tree cutting (like Mendha Lekha).

8 **How do forest dwellers contribute to the protection and conservation of forests?**

✔ ANSWER

- They hold traditional knowledge of herbs and plants and pass it down generations.
- They led forest protection movements — the Bishnoi Movement (Rajasthan, led by Amrita Devi) and the Chipko Movement (Uttarakhand).
- They take part in community forest management and eco-restoration (e.g. Mendha Lekha: regulating tree cutting, preventing fires).
- They protect sacred groves and follow rituals that show respect for nature.

9 **What is a forest fire? Prepare a plan for prevention and mitigation of forest fires.**

💡 SAMPLE ANSWER

A **forest fire** is an uncontrolled fire that spreads quickly through the dense vegetation of a forest, destroying trees, animals, homes and resources.

Prevention:

- Spread awareness; no campfires, cigarettes or burning of waste near forests.
- Clear dry leaves and make fire lines (gaps without vegetation).

- Watch towers and satellite alerts (e.g. School Bhuvan–NCERT Geoportal fire locations).

Mitigation:

- Report fires quickly to the forest department; trained local volunteer teams.
- Keep water, beaters and firefighting tools ready; evacuate people and animals safely.
- Replant trees after a fire.

🎉 **Chapter 2 done! Revise the answers once more tomorrow.**